Papers

Rare Gas Fluoride Lasers

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(Invited Paper)

Abstract—**In this article, the rare gas fluoride lasers are discussed in detail. There is a significant interest in these lasers because they are the most efficient visible/UV lasers to date. The dominant formation kinetics of KrF* and XeF* in both discharge and *e*-beam pumped lasers are presented. Because of the ionic upper level the formation processes are rapid and conditions can be chosen such that the branching ratios into the KrF* and XeF* levels from both the ionic and metastable levels are close to unity. With *e*-beam pumping of these lasers, a guide magnetic field enables the deposition of >90 percent of the beam energy into the optical volume. Discharge pumping has the potential of being more efficient than *e*-beam pumping; however, the key technical issues of discharge stability and metastable production efficiency have to be addressed. Stabilization of the discharge is possible if an external source of ionization is used. The quenching of the rare gas fluorides by two- or three-body processes has been carefully measured and analyzed. The three-body quenching of KrF* leads to the eventual formation of the excited triatomic Kr_2F^* which radiates in a broad band centered at 410 nm. It has also been determined that ionic and excited state absorption in the active media is large enough to impact the extraction efficiency of these lasers. In the case of KrF*, the dominant absorbing species are F_2, F^-, and Kr_2F^*, while Xe_2^+ appears to be the dominant absorber in XeF*. Finally, we have also investigated experimentally and theoretically the effects of the vibrational relaxation in the upper level and finite lifetime of the lower level on XeF* laser performance.**

I. INTRODUCTION

RARE gas fluoride exciplexes are the most efficient visible/UV lasers to date. The rare gas monohalide lasers have been actively investigated since their emission spectra were reported in 1975 by Velazco and Setser [1] and by Golde and Thrush [2]. Table I lists the rare gas halide exciplexes that have been made to lase together with spectroscopic data for the lasing transition. XeBr* was the first of this class to lase. This molecule was produced first by relativistic electron beam pumping by Searles and Hart [3]. Shortly after that Ewing and Brau [4], [5] obtained lasing action in XeF*, XeCl*, and KrF*. Recently a new family of lasers, the mercury monohalides, has been demonstrated by Parks [6]. The possibility of lasing the mercury monohalides was first predicted by Ewing [7]. All the exciplexes listed in Table I have been made to lase by relativistic *e*-beam pumping, and many of them have been made to lase in *e*-beam controlled discharges and UV preionized discharges. The best results obtained to date by pure *e*-beam pumping [8] of a KrF* laser are 12 J/l extracted at 9 percent intrinsic efficiency.

Pumping of the rare gas monohalide lasers with an *e*-beam controlled discharge was first demonstrated by Mangano and Jacob [9]. In *e*-beam controlled discharge pumping, a discharge voltage is applied across the mixture which is ionized by an *e*-beam. This method has the potential of being more efficient than pure *e*-beam pumping. However, realization of high efficiency depends on tradeoffs relating discharge stability, exciplex production efficiency, and the discharge enhancement ratio (discharge power/*e*-beam power). The best results to date achieved with this method of pumping in KrF* is 10 J/l at an intrinsic efficiency of 9.5 percent, [10]. The third method of exciting these lasers is the UV-preionized discharge [11], [12]. This is the simplest system to build; however, the efficiency is limited to about 1 percent. In this method, ultraviolet light is used to preionize the medium prior to initiation of the avalanche discharge. Avalanche discharge pumping is more compatible with higher pulse-repetition rates than other methods and should find wide use in applications in which high rep rate is more important than efficiency. Potential applications include isotope separation and other photochemical processes which require pulses of 1 J or less. Avalanche pumping lacks the control of *e*-beam ionized discharges and so the power and pulse lengths attainable are limited, as will be discussed in the text. The controlled-discharge technique gives much higher energy per pulse and has the potential of being scaled to higher average power than the avalanche approach. This is because in a stable discharge, longer laser pulses with higher efficiency are possible.

To build an efficient laser the following criteria have to be met: 1) the formation efficiency of the upper laser level must be high; the formation efficiency is the product of the branching ratio (from the ionic or metastable states to the upper laser level) and the quantum efficiency; 2) the quenching of the upper laser level should not be extremely rapid; 3) the absorption of

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TABLE I
CANDIDATE LASER MOLECULES SPECTROSCOPIC DATA FOR THE $B \longrightarrow X$ LASING TRANSITIONS OF THE RARE GAS HALIDES. IN THIS TABLE, λ = TRANSITION WAVELENGTH, r = EQUILIBRIUM INTERNUCLEAR SEPARATION, ω_e = FUNDAMENTAL VIBRATIONAL FREQUENCY OF THE B STATE, σ = STIMULATED EMISSION CROSS SECTION, AND τ = RADIATIVE LIFETIME OF THE B STATE

EXCIPLEX	$\lambda(B \rightarrow X)$ (nm)	r (Å)	ω_e (cm^{-1})	$\sigma\tau(B \rightarrow X)$ ($Å^2$ – ns)	$\tau(B \rightarrow X)$ (ns)
ArF^*	193[a]	2.2[f]	–	12[h]	4.2[j], 6[k]
KrF^*	249[b]	2.3[f]	310[b]	17[b]	6.7[j], 9[l]
$KrCl^*$	222[c]	2.8[f]	210[g]	–	–
XeF^*	351[d]	2.4[f]	309[d]	64[i]	12[j], 18.8[m]
$XeCl^*$	308[e]	2.9[f]	194[e]	50[f]	11[j]
$XeBr^*$	282[b]	3.1[f]	120[b]	26[b]	12[j], 17.5[n]

a. REF 70 f. REF 27 k. REF 77
b. REF 65 g. REF 74 l. REF 78
c. REF 71 h. REF 75 m. REF 79
d. REF 72 i. REF 14 n. REF 80
e. REF 73 j. REF 76

the active media should be small to enable efficient optical extraction; and 4) the lower laser level should be removed rapidly. To date, the lasers that have shown the most promise are the rare gas fluorides which have high formation efficiencies, both by *e*-beam [13], [14] and *e*-beam controlled discharge pumping [15]. In this paper we will discuss in detail the physics of the KrF* and XeF* lasers which are typical of efficient ionically bound exciplex lasers. The main difference between KrF* and XeF* is that KrF* has an unbound lower level while the lower level of XeF* is bound by about 0.15 eV. The effects of the bound lower level on the laser performance will be discussed subsequently. We will show the extent to which the above criteria are satisfied by these lasers.

In Fig. 1 we see a schematic potential energy diagram of a rare gas halide exciplex. There are two noteworthy features: 1) the upper level correlates with a $R^+ + X^-$ ion pair where R^+ denotes rare gas ion and X^- a halogen ion; and 2) the lower level is in most cases unbound. The laser transitions are characterized by large stimulated emission cross sections (1-5 $Å^2$) and short radiative lifetimes (4-20 ns). The ionic nature of the upper level allows for rapid formation via both the ionic [13], [14] and metastable channels [1]. In most rare gas fluorides there are only a few curve crossings with excited molecular levels. As a result, the probability of interception is small and the formation efficiency is high. The ionic channel is usually dominant in pure *e*-beam pumping (see Fig. 1), while in *e*-beam controlled discharge pumping, most of the formation proceeds via the metastable channel. In the case of KrF*, the repulsive potential between Kr and F ensures a negligible lower level population, automatically satisfying the fourth criteria for an efficient laser mentioned previously.

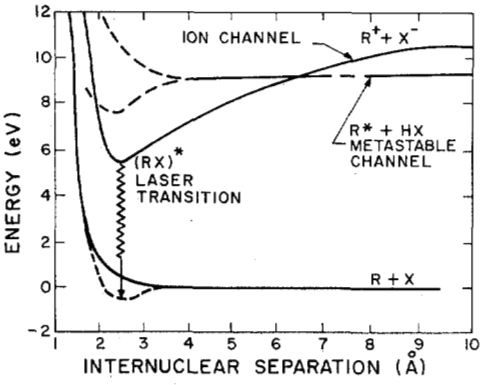


Fig. 1. Typical potential energy diagram of a rare gas fluoride.

II. FORMATION KINETICS

A. E-Beam Pumped KrF*

Typical mixtures contain about 90 percent Ar, <10 percent Kr, and ≤0.5 percent F_2 at a total pressure of 1-3 atm. Fig. 2 shows schematically the dominant formation kinetics. Since the mixture contains mainly Ar, about 55 percent of the *e*-beam power deposited results in argon ion-electron pairs. Approximately 10 percent of the power deposited results in Ar metastable formation. For typical laser conditions (i.e., *e*-beam current densities of 5-10 A/cm^2 and gas mixtures quoted above), the dominant loss of secondary electrons is rapid dissociative attachment with F_2 resulting in F^- formation. At pressures below 1 atm, the Ar^+ and F^- recombine to form ArF*. The ArF* can radiate at 193 nm to form ground state Ar and F atoms with a calculated radiative lifetime of about 4 ns [16]. However, for mixtures containing >5 percent

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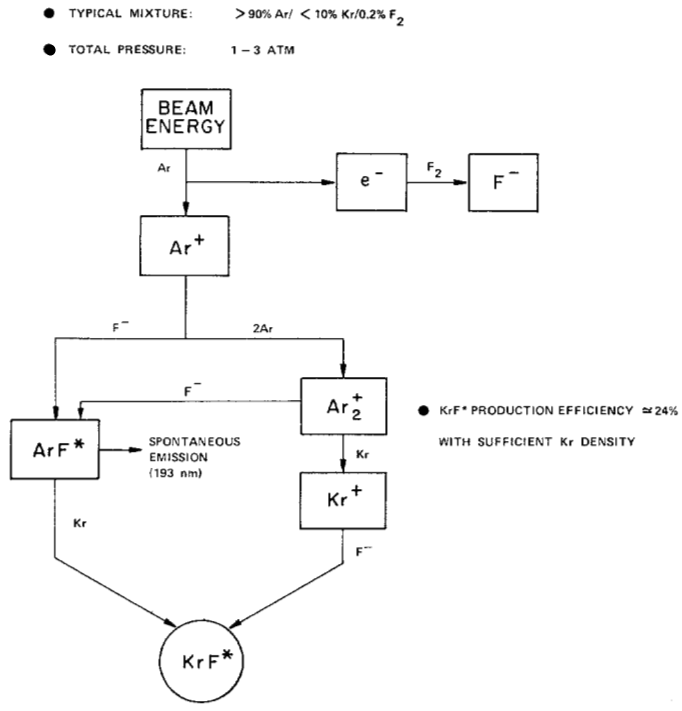


Fig. 2. Dominant formation kinetics of e-beam pumped KrF*.

krypton [13] and mixture pressures $>\frac{1}{2}$ atm, the Kr will exchange efficiently with the ArF* to form KrF*. At pressures above 1 atm the Ar^+ and Ar react predominantly to form Ar_2^+ with a three-body rate constant of 2.5×10^{-31} cm^6/s [17]. The Ar_2^+ then undergoes charge transfer with Kr to form Kr^+ with a rate constant of 7.5×10^{-10} cm^3/s [18]. For lean Kr mixtures, the Kr^+ will combine with the F^- to form KrF*. As the Kr number density is increased (or the total mixture pressure), molecular krypton ions are formed. The Kr_2^+ have also been shown to recombine with F^- to form mainly KrF* [13]. The KrF* can spontaneously decay with a lifetime of 6.5 ns [19] or can be collisionally quenched. The quenching processes will be discussed subsequently.

Before we discuss the formation of XeF* a comment should be made concerning the use of NF_3 as a fluorine donor in KrF* laser mixtures. NF_3 is an attractive donor because unlike F_2 it does not absorb the KrF* radiation. However, higher fluorescence and laser efficiencies are achieved with F_2 than with NF_3 [20]. This occurs because Kr^+ can effectively charge transfer to NF_3 to form NF_3^+ [21]. Such a reaction is of course a loss process. Interestingly enough $Xe^+(^2P_{3/2})$ does not have sufficient energy to ionize NF_3 [22]. In Ne rich mixtures XeF* is formed mainly via $Xe^+(^2P_{3/2})$ as will be discussed subsequently. These facts together with the fact that NF_3 does not photoabsorb at 350 nm make NF_3 a useful fluorine donor in the XeF laser.

B. E-Beam Pumped XeF

Typical mixtures contain $\leqslant$0.5 percent NF_3, $\leqslant$1 percent Xe and the remainder is Ar or Ne. The formation kinetics in Ar-rich XeF* mixtures are similar to KrF* which have been discussed previously. In XeF* laser mixtures, NF_3 is used instead of F_2. The main advantage of using NF_3 is that it does not absorb the XeF* laser radiation [23]. The details of the formation processes with Ne are not as well established. However, experimental results indicate that the branching into XeF* in Ne-diluted mixtures is the same, within experimental error, as in Ar-diluted mixtures. This conclusion is reached by comparing the fluorescence intensity of $Ne/Xe/F_2$ mixtures with that of similar $Ar/Xe/F_2$ mixtures (from $\frac{1}{2}$–4 atm) [24]. In making these comparisons, the difference in the e-beam stopping power, the energy required to create an electron/ion pair, and the different quenching rates of XeF* for the different mixtures, were taken into account.

The fluorescence spectra for typical $Ar/Xe/F_2$ and $Ne/Xe/F_2$ mixtures are shown in Fig. 3. These data were taken at a total mixture pressure of 3 atm and contained 0.3 percent F_2 and 0.5 percent of Xe. From the spectra in Fig. 3, one sees two differences between the Ne and Ar-diluted mixtures, 1) the $XeF^*(D \rightarrow X)$ emission at 260 nm is present only in the excited $Ar/Xe/F_2$ mixture, and 2) the $(B \rightarrow X)$ emission in the Ne-rich mixture is broader than in the Ar-rich mixture. The D state correlates with $Xe^+(^2P_{1/2})$ while the B state correlates with $Xe^+(^2P_{3/2})$ [25]. These differences were present in all spectra obtained in the pressure range between 0.5–3 atm. The greatly reduced intensity of the $D \rightarrow X$ emission in Ne-rich mixtures is because the $Xe^+(^2P_{1/2})$ is not created in these mixtures, even though Ne_2^+ has sufficient energy to charge transfer to $Xe^+(^2P_{1/2})$ and $Xe^+(^2P_{3/2})$. This implies that Ne_2^+ charge transfer to Xe^+ is not a dominant XeF* formation process [26]. Also, NeF* has 11.6 eV of energy [27] which is 3 eV larger than the $Xe^+ + F^-$ energy at infinite separation. So there is no curve crossing and it is unlikely that the Ne atom in NeF* will be displaced by Xe to form XeF*.

A possible formation chain that is consistent with experimental observations is shown in Fig. 4. Since Ne is the major species in the mixture, NeF* will be formed by reaction of Ne^+ and Ne_2^+ with F^- [13], [14]. Once NeF* is formed, it can spontaneously radiate or predissociate to F* and Ne. The predissociation of NeF* is expected [29] because atomic fluorine has an ensemble of levels (^4P and ^2P) about 1 eV above the bottom of the NeF* potential well. NeF* can also collisionally excite Xe to high lying excited levels which will eventually form XeF*. The F* which is produced by the NeF* predissociation can radiate, Penning ionize Xe, or form F_2^* by reaction with F_2. F* is energetic enough to ionize Xe to the $^2P_{3/2}$ level only. Also, the radiated photons from F* can photoionize Xe. The cross section for this process is 6×10^{-17} cm^2 [30], resulting in a photoabsorption length of 1.5 mm in a 0.5 percent Xe mixture at a total pressure of 1 atm. The F_2^* can react with Xe in a displacement reaction to form XeF*. All the above kinetic processes will produce XeF(B) and not XeF(D) [24]. Hence, the $D \rightarrow X$ emission at 260 nm is not observed in Ne-rich mixtures as shown in Fig. 3.

It is interesting to note that in $Ne/Xe/NF_3$ mixtures the $D \rightarrow X$ emission at 260 nm is stronger than in similar $Ne/Xe/F_2$ mixtures. The reason for the difference is that NF_3 and Xe react with Ne metastables (Ne*) with almost the same rate constant [28]. Most of the metastables are formed directly by the e-beam as mentioned in Section I. As the density of

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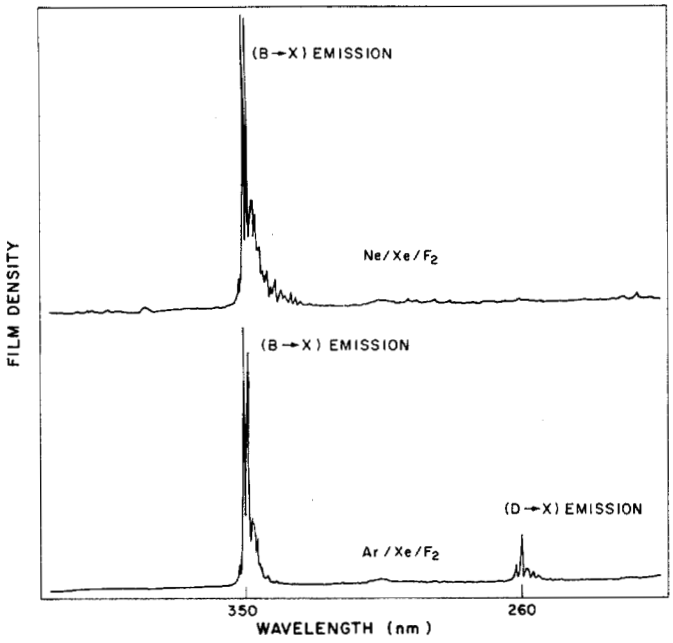


Fig. 3. Comparison between emission spectra of $Ar/Xe/F_2$ and $Ne/Xe/F_2$ mixtures. The mixtures contain 0.3 percent F_2, 0.5 percent Xe at a total pressure of 3 atm.

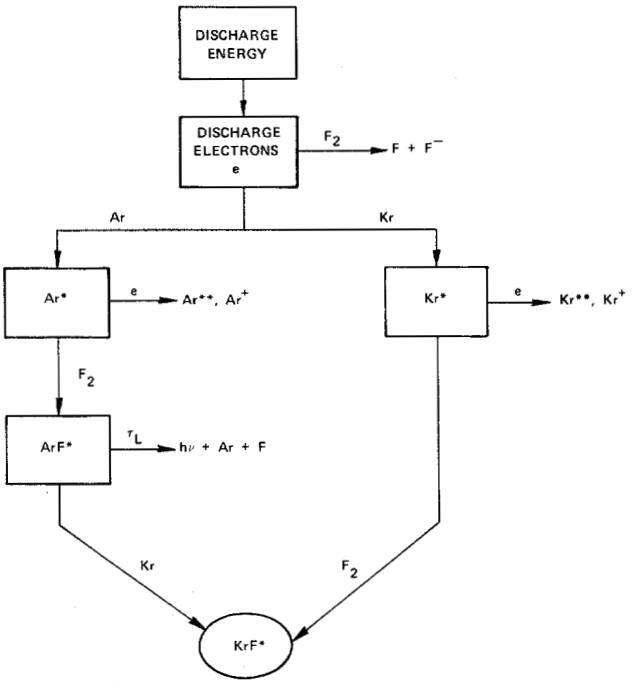


Fig. 5. Dominant formation kinetics of discharge pumped KrF*.

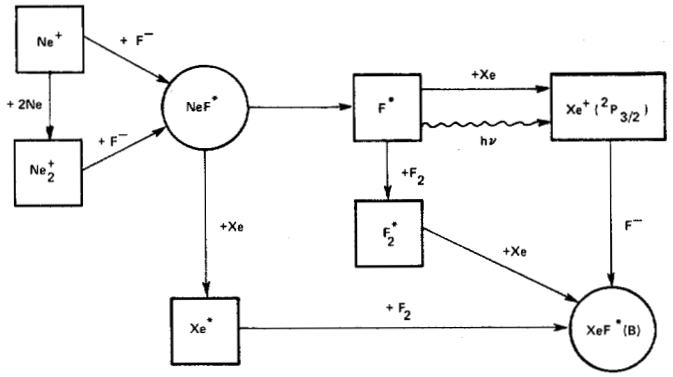


Fig. 4. Possible XeF* formation processes in *e*-beam pumped $Ne/Xe/F_2$ mixtures.

Xe is two to five times the halogen donor, most of the Ne* in mixtures containing NF_3 will lose its energy via Xe. Ne* has sufficient energy to Penning ionize Xe to both $Xe^+(^2P_{1/2})$ and $Xe^+(^2P_{3/2})$ states. The reaction rate constant of Ne* with F_2 is about five times faster than the rate constants of Ne* with Xe [28]. Hence in $Ne/Xe/F_2$ most of the Ne* will react with F_2 rather than Xe.

With an Ne buffer, the emission on the $B \rightarrow X$ band of XeF* is broader than with an Ar buffer indicating that the vibrational relaxation in the XeF(B) manifold by Ne is slower than that by Ar. Since Ne is lighter than Ar, relaxation by V-T transfer would be expected to proceed more rapidly in Ne-rich mixtures. So it appears that V-T energy transfer is not the dominant process. A possible explanation is that Ar and Xe can form a molecular ion $ArXe^+$ whereas $NeXe^+$ is more weakly bound or unbound [31]. Hence, an intermediate complex ArXeF* can be formed which may result in a more rapid vibrational relaxation of XeF* [24]. The formation of such complexes may also explain why Ar quenches XeF* more rapidly than Ne.

C. Discharge Pumped KrF*

The gas mixtures for discharge pumping and *e*-beam pumping of KrF* are similar. For efficient discharge pumping the formation should proceed via the metastable channel. Fig. 5 shows schematically the dominant formation kinetics.

In discharge pumping the secondary electrons produced by an *e*-beam or by UV photons gain enough energy in an applied electric field to excite the argon and krypton metastables. The argon metastables, Ar*, produce ArF* with a branching ratio of about 0.6 [32] via the harpoon reaction with F_2($Ar^* + F_2 \rightarrow ArF^* + F$). The rate constant for this reaction has been measured by Velazco, Kolts, and Setser to be 7.8×10^{-10} cm^3/s [33]. As we have discussed previously, the ArF* can either radiate or the Ar atom can be displaced by the Kr to form KrF*. The krypton metastable Kr* can form KrF* directly, and with a branching ratio of unity, via the harpoon reaction with F_2 [33]. Because of the lower energy required to excite Kr* and the unit branching ratio to form KrF*, the formation efficiency will be larger if most or all the discharge energy goes into Kr*. The fraction of discharge energy that is deposited into Kr* will depend on the percentage of Kr and applied electric field as shown in Fig. 6. Clearly, the higher the fraction of Kr, the larger the formation efficiency. However, as we will show subsequently, Kr quenches KrF* rapidly and hence the fraction of Kr needs to be kept small. For 10 percent Kr, the KrF* formation efficiency by discharge pumping under stable and controlled conditions can be as high as 30–35 percent, as shown in Fig. 7.

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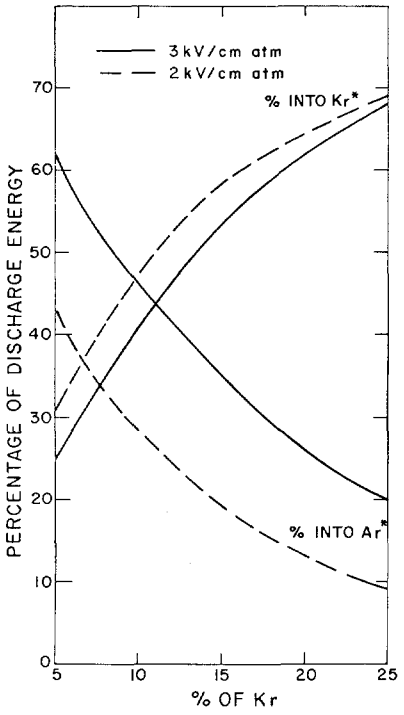


Fig. 6. Percentage of discharge energy into Ar* and Kr* as a function of fractional Kr density for Ar/Kr mixtures.

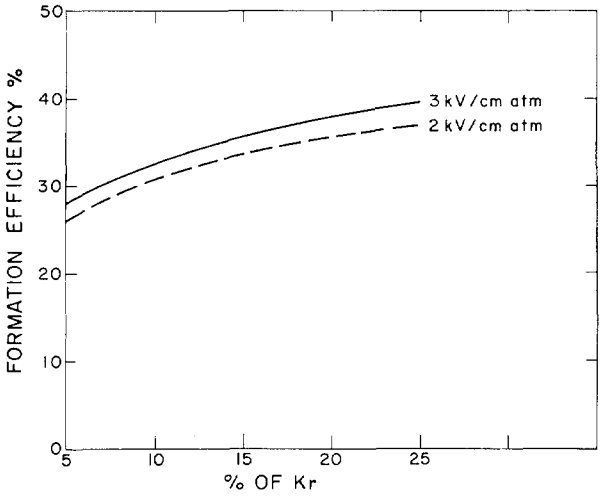


Fig. 7. Formation efficiency of KrF* as a function of fractional Kr density.

D. Discharge Pumped XeF*

We will first discuss the formation kinetics in $Ar/Xe/NF_3$ mixtures. The secondary electrons produced by an external source of ionization gain energy in the applied electric field to enable the excitation of the xenon and argon metastables

$$e_s + R \longrightarrow R^* + e_s. \tag{1}$$

The fraction of discharge power that initially goes into producing Ar* and Xe* is shown in Fig. 8 for electric fields of 2 and 3 kV/cm · atm. From this figure it is apparent that for a 0.5 percent Xe mixture most of the discharge power goes into Ar* formation. The power that goes into Xe* formation rapidly results in XeF* with a unit branching ratio

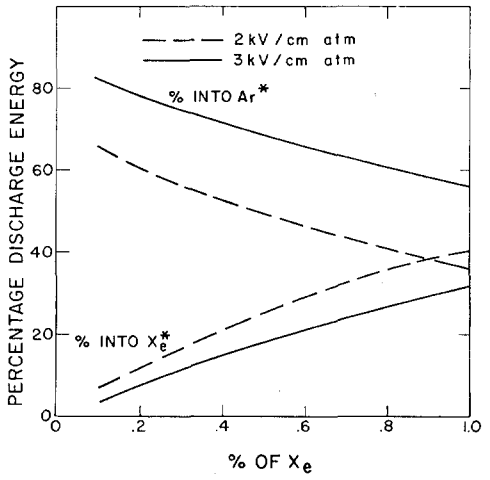


Fig. 8. Percentage of discharge energy into Ar* and Xe* for Ar/Xe mixtures as a function of fractional Xe densities.

via the harpoon reaction with NF_3 [33]. There are three principal reactions by which the argon metastables are lost:

$$Ar^* + Xe \longrightarrow Xe^{**} + Ar \tag{2}$$

$$Ar^* + 2Ar \longrightarrow Ar_2^* + Ar \tag{3}$$

$$Ar^* + NF_3 \longrightarrow ArF^* + NF_2. \tag{4}$$

The rate constants for reactions (2) have been measured to be 1.8×10^{-10} and 3×10^{-10} cm^3/s for Ar($^3P_2$) and Ar($^3P_0$), respectively [34]. Reactions (3) have three-body rate constants of 1.7×10^{-32} cm^6/s for Ar($^3P_1$), 0.9×10^{-32} cm^6/s for Ar($^1P_1$) and 1.6×10^{-32} cm^6/s for Ar($^3P_2$) [35]. Electron mixing of the Ar excited states only marginally affects the loss rates of Ar* by reactions (2) and (3). So we will assume a mean rate constant of 2×10^{-10} cm^3/s for reaction (2) and 10^{-32} cm^3/s for reaction (3). The argon metastables react with NF_3 to form ArF* with a rate constant of 1.4×10^{-10} cm^6/s [33]. At a mixture pressure of 2 atm with 0.5 percent Xe and 0.1 percent NF_3, the rates for reactions (2), (3), and (4) are 4×10^7, 0.6×10^7, and 0.7×10^7 s^{-1}. So about 15 percent of the Ar* is channeled into ArF*. Most of the energy in ArF* will produce XeF* by the following reaction

$$ArF^* + Xe \longrightarrow XeF^* + Ar. \tag{5}$$

The remainder of the energy is presumably radiated on the ArF* bands. Since reaction (5) proceeds with a rate constant of 1.6×10^{-9} cm^3/s [36], 65 percent of the ArF* is channeled into XeF*.

The argon excimers Ar_2^* will be deactivated at least as rapidly as Ar* by Xe to form highly excited xenon states Xe**. This means that 70-80 percent of the Ar_2^* will end up as Xe**. The remainder of the Ar_2^* will be radiated. So we can conclude that the argon metastables transfer their energy to the xenon metastables with about an efficiency of 80 percent. The Xe** thus formed will react with the NF_3 to form XeF*. The branching ratio and rate constant for this reaction are not known. However, from discharge data, a reaction rate constant of 4×10^{-10} cm^3/s and a branching ratio of 0.7 have been inferred [37].

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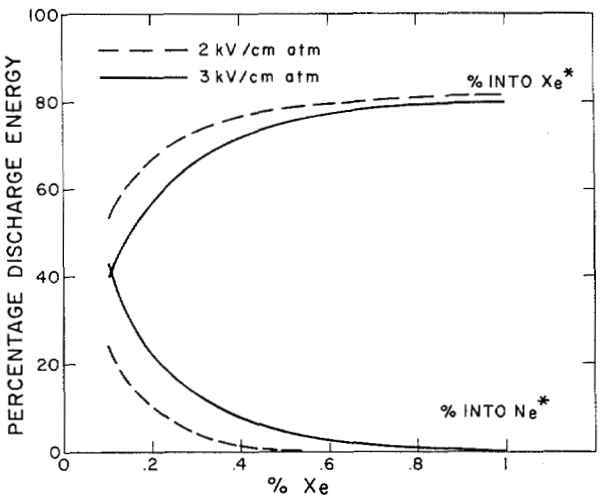


Fig. 9. Percentage of discharge energy into Xe* and Ne* for Ne/Xe mixtures as a function of fractional Xe densities.

When the Ar buffer is replaced by Ne, most of the discharge energy will result in Xe* formation as shown in Fig. 9. The Xe* will produce XeF* via a harpoon reaction with NF_3 [33]. Hence for typical laser mixtures containing Ne, the only factor influencing the XeF* formation efficiency is the Xe* production efficiency.

III. Quenching Kinetics of the Rare Gas Fluorides

Once the excited rare gas fluoride is formed it can radiatively decay, be quenched by two- and three-body collisions with F_2 or NF_3 and the rare gases, or under lasing conditions be stimulated by cavity flux. For efficient laser power extraction, the probability of stimulated emission should be much larger than the combined probability of decay by radiative and collisional processes. In the absence of a lower level the ratio of these probabilities is given by ϕ_c/ϕ_s, where ϕ_c is the cavity flux and ϕ_s is the saturation flux, which is defined by

$$\phi_s = \frac{h\nu}{\sigma_s}\left[\frac{1}{\tau_r} + K_x N_x + \sum_R K_R N_R + \sum_{R,R'} K_{RR'} N_R N_{R'}\right] \quad (6)$$

where $h\nu$ = photon energy; σ_s = stimulated cross section; τ_r = the exciplex radiative lifetime; K_x is the two-body quenching rate constant by the halogen donor; K_R and $K_{RR'}$ are the two- and three-body quenching rate constants by the rare gases. N_x is the number density of the halogen donor and N_R and $N_{R'}$ are the number densities of the different rare gases.

A. Dominant Quenching Processes of KrF*

The different measured quenching rate constants of KrF* are listed in Table II. These rate constants were determined by observing the KrF* $B^2\Sigma_{1/2} \rightarrow X^2\Sigma_{1/2}$ radiation as a function of the partial pressures of the various gas constituents [13], [38]. It is of interest to note that the three-body quenching of KrF* by Ar and Kr leads to the eventual formation of Kr_2F^* by the following reactions:

$$KrF^* + 2Ar \longrightarrow Ar\,KrF^* + Ar \quad (7)$$

followed by

$$Kr + Ar\,KrF^* \longrightarrow Kr_2F^* + Ar \quad (8)$$

or directly by

$$KrF^* + Kr + M \longrightarrow Kr_2F^* + M. \quad (9)$$

The appearance of Kr_2F^* at high pressures is demonstrated in Fig. 10. In this figure, the spontaneous emission spectra from *e*-beam excited mixtures were obtained at mixture pressures of 0.5, 2, and 4 atm. The uncalibrated intensity scale is approximately logarithmic. At 0.5 atm essentially all of the radiation from the mixture is contained in the KrF* $B^2\Sigma_{1/2} \rightarrow X^2\Sigma_{1/2}$ band at 248 nm. However, two other broadbands, containing much less energy, are observable. The first is a Kr_2F^* band ($^2B_2 \rightarrow A_1$) centered at 415 nm. The other broadband, centered roughly at 270-280 nm, is most likely a combination of radiation from the $^2\Sigma \rightarrow {}^2\Pi$ of KrF* and perhaps radiation from the excited triatomics Ar_2F^* and Ar KrF*. The spectrum obtained at a total mixture pressure of 4 atm indicates that, compared with the 0.5 atm spectrum, essentially the same energy is contained in the KrF* $B^2\Sigma_{1/2} \rightarrow X^2\Sigma_{1/2}$ band, although the energy deposited increased by a factor of ≈ 8. This spectrum indicates that most of the additional energy deposited by the *e*-beam was channeled into Kr_2F^* by the quenching process discussed above [13].

B. Dominant Quenching Processes of XeF*

The two- and three-body quenching rate constants of XeF* by F_2, NF_3, and the rare gases are listed in Table III. The measurements of these rate constants have been discussed in [14], [24], and [39]. From this table it is clear that the quenching of XeF* by Ne is significantly slower than by Ar. This is one of the reasons that Ne is a better diluent than Ar in the XeF* laser. One main difference between the quenching process of XeF* and KrF* is that two-body quenching by the rare gases plays a more important role in XeF* than in KrF*. This is especially true of the quenching of XeF* by Xe [39] as compared to the quenching of KrF* by Kr.

IV. Pumping Considerations

As we have mentioned in the Introduction, the three main pumping methods are: 1) direct *e*-beam pumping; 2) *e*-beam controlled discharge pumping; and 3) UV preionized (avalanche) discharge pumping. In this section we will discuss these pumping methods and aspects peculiar to each.

A. E-Beam Pumping

The use of high energy electrons has greatly advanced the technology of high power lasers. Fast electrons were first used as a source of ionization in high pressure CO_2 lasers [40]. They were subsequently used for the direct pumping of rare gas excimers [41] and rare gas monohalide lasers [3], [5]. In fact, all the excimer and exciplex lasers were first pumped by high energy *e*-beams. More recently they have been used as a partial source of ionization in discharge pumping of the KrF* and XeF* lasers [9]. One of the scaling

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TABLE II
DOMINANT QUENCHING PROCESSES OF KrF*

REACTION	$k\tau_R(KrF^*)$	$k(\tau_R = 6.5$ nsec)
$KrF^* + F_2 \rightarrow$ PRODUCTS	5×10^{-18} cm^3	7.8×10^{-10} cm^3 sec^{-1}
$KrF^* + 2Kr \rightarrow Kr_2F^* + Kr$	4.4×10^{-39} cm^6	6.7×10^{-31} cm^6 sec^{-1}
$KrF^* + Kr \rightarrow$ PRODUCTS	$\leq 1.1 \times 10^{-20}$ cm^3	
$KrF^* + Kr + Ar \rightarrow Kr_2F^* + Ar$	4.2×10^{-39} cm^6	6.5×10^{-31} cm^6 sec^{-1}
$KrF^* + 2Ar \rightarrow$ PRODUCTS	4.6×10^{-40} cm^6	7×10^{-32} cm^6 sec^{-1}

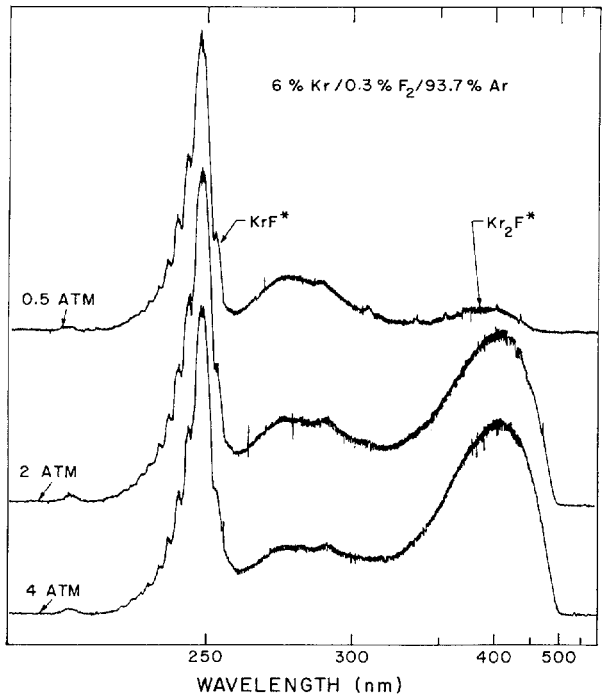


Fig. 10. Spectra from a KrF* laser mixture at various total pressures.

TABLE III
DOMINANT XeF* QUENCHING PROCESSES

REACTION	$k\tau_r$	$k(\tau_r = 16$ ns)
$XeF^* + F_2$	5.3×10^{-18} cm^3	3×10^{-10} cm^3/sec
$XeF^* + NF_3$	2.8×10^{-19} cm^3	1.7×10^{-11} cm^3/sec
$XeF^* + Xe$	4.5×10^{-19} cm^3	2.9×10^{-11} cm^3/sec
$XeF^* + Ne$	NEGLIGIBLE	
$XeF^* + Xe + Ne$	1.23×10^{-38} cm^6	7.7×10^{-31} cm^6/sec
$XeF^* + 2Ne$	4.32×10^{-41} cm^6	2.7×10^{-33} cm^6/sec
$XeF^* + Ar$	1.28×10^{-20} cm^3	8×10^{-13} cm^3/sec
$XeF^* + 2Ar$	2.4×10^{-40} cm^6	1.5×10^{-32}cm^6/sec
$XeF^* + Xe + Ar$	4.8×10^{-39} cm^6	3×10^{-31} cm^6/sec

limitations of these lasers is the magnetic field resulting from the discharge current and self-magnetic field of the beam that causes the beam to pinch. Boyer, Henderson, and Morse [42] have demonstrated the effects of the discharge magnetic field on the fast electrons by Monte-Carlo simulations. These authors have also shown that a guide magnetic field can reduce and eliminate the effects of pinching. Mangano [43] has recently shown the effects of *e*-beam pinching resulting from a self-magnetic field. He has also demonstrated that a guide magnetic field eliminates this effect and also guides the *e*-beam against scattering by the foil and the high pressure gas. In fact, with the application of a guide magnetic field, direct pumped KrF* laser output energy can be increased by a factor of four in typical devices [10]. The increased laser performance is a direct result of more efficient coupling of the *e*-beam into the laser mixture.

Electron beams are widely used because of their availability and their ability to deposit relatively large power densities uniformly into large volumes of laser mixture. The pump power density P required to obtain lasing action for an exciplex with a repulsive lower level is given by

$$P \gtrsim \frac{gE}{\eta\tau\sigma_s L} \tag{10}$$

where g is the required single pass gain, τ the exciplex lifetime, σ_s the stimulated cross section, L the length of the active medium, E the energy of the upper level, and η the efficiency of exciting the upper level. So for typical rare gas fluoride laser conditions (a single pass gain of 20 percent, a lifetime of 10 ns, efficiency of 10 percent, length of 20 cm, $\sigma_s = 10^{-16}$ cm^2 and $E \approx 5$ eV)

$$P \gtrsim 10^5 \text{ W/cm}^3. \tag{11}$$

Such high pump power densities are easily obtained by *e*-beams.

In this section, we discuss electron scattering in gases by assuming that the fast electrons can be described by a current and number density. Such a simplification is not valid for a well collimated beam. However, the high energy electrons have first to pass through a foil; if the foil thickness is greater than $\frac{2}{5}$ of the transport mean free path, then the fast electrons, after traversing the foil, are fairly diffuse in angle

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[44], [45]. Further by a superposition of two isotropic sources, a well collimated beam may be treated by the diffusion theory. For a diffuse beam, in the absence of electric and magnetic fields, one can use the "age" theory [44]. In addition, the age theory has been extended to include the effects of the discharge electric field [46]. The importance of the electric field can be assessed by considering the dimensionless parameter γ, which is the energy gained per transport mean free path divided by the electron energy. Typically, in an *e*-beam controlled discharge $\gamma \approx 0.1$–0.3 and the effect of the electric field on fast electrons is small, so we will ignore it.

Besides the effect of imposed electric fields it is necessary to consider the effect of both self-magnetic fields and imposed magnetic fields. Diffusion in the presence of a magnetic field can generally be separated into two directions: diffusion along, and perpendicular to, the field direction. This analysis is similar to diffusion of a plasma in a magnetic field. The diffusion across the magnetic field is restricted because the electrons will tend to gyrate around magnetic field lines. In fact, the diffusion across the field lines will be reduced by a factor $(1 + \lambda^2/r_l^2)^{-1}$ where λ is the transport mean free path and r_l is the Larmor radius [47]. However, because of the energy dependence of λ and r_l, the resulting diffusion perpendicular to the magnetic field is a complicated function of magnetic field strength B and the electron energy. In the limit of large B, the diffusion perpendicular to field lines turns out to be inversely proportional to the magnetic field strength [47]. Reasonable magnetic field strengths can in fact be imposed to good advantage. For typical rare gas exciplex laser mixtures and pressures λ/r_l is $\gg 1$ for magnetic fields of 1 kG.

In the presence of a strong magnetic field, only the diffusion of electrons in the direction of the field need be considered. The half space solution for this case simplifies to [46]

$$N_h(x,t) = N(x - 0.4\lambda, t) - N(x + 1.82\lambda, t)$$
$$= \frac{1}{\chi}\sqrt{t}\left[\exp - \frac{(x - 0.4\lambda)^2}{4t} - \exp - \frac{(x + 1.82\lambda)^2}{4t}\right] \quad (12)$$

where N_h = energy deposition, x = distance along the beam, and t = electron age [46]. To obtain the energy deposition one has to integrate over all t. Fig. 11 shows the solution for a mixture containing 95.7 percent Ar, 4 percent Kr, and 0.3 percent F_2 at a total pressure of 1.7 atm. In this calculation we have also allowed for a 2 mil Kapton foil.

The most efficient way to couple *e*-beam energy into the laser mixture is to use two *e*-beams in a face to face configuration. The energy deposition resulting from such a configuration is shown in Fig. 12. Using two *e*-beams together with a magnetic field, 90 percent of the fast electron energy can be coupled into the laser mixture with a variation in the energy deposition of ±10 percent across the laser cavity in the beam direction.

B. Discharge Pumping

The two most commonly used discharge pumping methods are *e*-beam controlled discharge pumping and UV preionized, avalanche discharge pumping. With *e*-beam controlled discharge pumping, most, but not all, the discharge electrons are produced by the high energy *e*-beam. The discharge, which provides most of the pump power, is then run in the ionization produced by the fast electrons. In UV preionized discharge pumping, the laser medium is preionized by UV radiation.

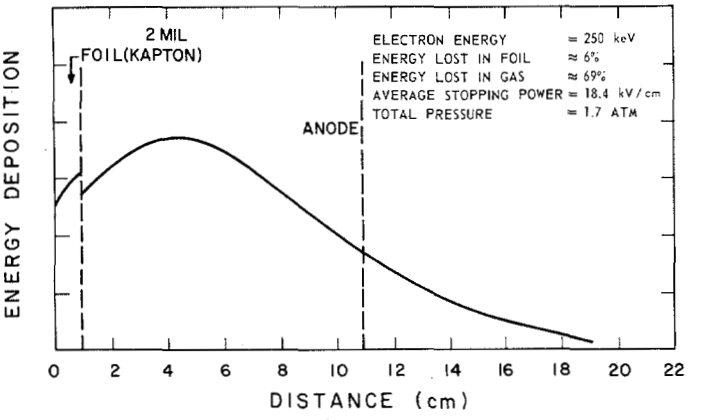


Fig. 11. *E*-beam energy deposition in a $Ar/Kr/F_2$ mixture at total pressure of 1.7 atm.

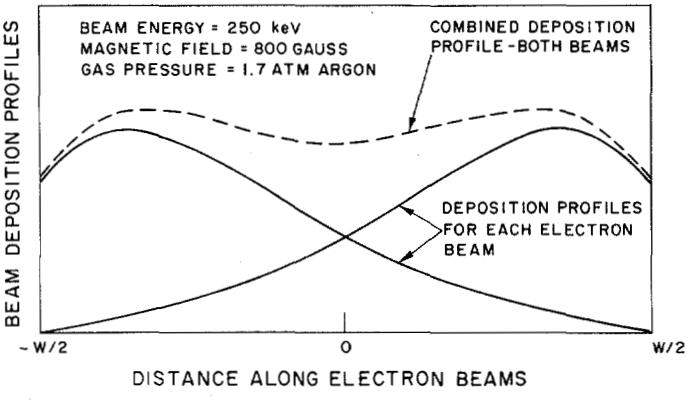


Fig. 12. Predicted *e*-beam deposition when two *e*-beams are used.

1) Metastable Production Efficiency

The discharge physics is strongly affected by electron impact excitation and ionization of the rare gas metastables. To model these effects we have treated the krypton metastable as rubidium and the argon metastable as potassium. This analogy has been used successfully in predicting the emission spectra of the excited rare gas monohalides and is justified physically by the atomic similarity between rare gas metastables and the alkalis [27]. Some of the electron impact cross sections used in our model are shown in Fig. 13. The cross section for excitation from the 5*s* configuration to the 5*p* configuration in Rb [48] (Kr*) has a peak value of 75 Å^2 at 8 eV. Also shown is the ionization cross section of Rb [49] (Kr*) and the excitation [50] and ionization [51] cross sections of ground state Ar. From Fig. 13 it is clear that the peak value of the metastable Ar excitation cross section is 30 times the peak value of the excitation cross section of Ar from the ground state. More important, however, is the fact that most of the electrons can excite the 5*s* to 5*p* transitions which

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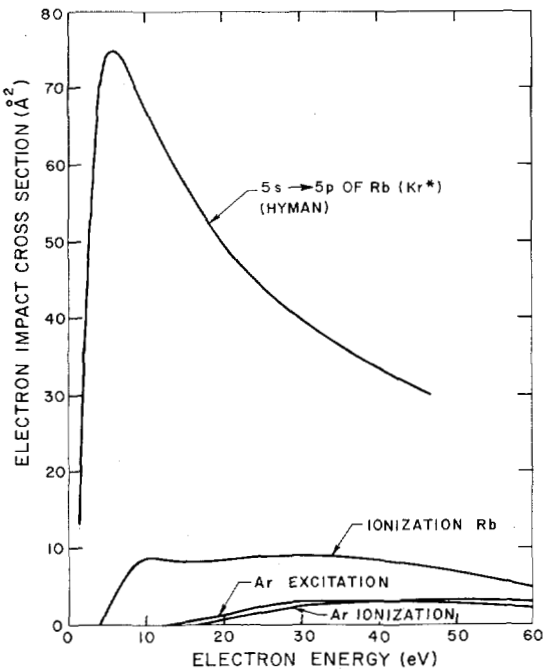


Fig. 13. Electron impact cross section for Rb(Kr*) and Ar as a function of electron energy.

have a threshold of 1.6 eV, resulting in a cooling of the electrons whereas only the high energy tail of the electron energy distribution can produce metastables from the ground state. We have put these cross sections into a computer code which solves the Boltzmann electron transport equation. This Boltzmann code takes the cross section data and the electric field and calculates self-consistently the electron energy distribution and the partitioning of discharge energy among the various excited states and ionization. The predictions of the code are shown in Figs. 14-16 [52]. Fig. 14 shows the percentage of energy that goes into producing Kr* as a function of the fractional metastable population $Kr^*/(Kr + Ar)$ for electric fields of 2-6 kV/cm · atm. It is apparent from Fig. 14 that the efficiency of producing the metastables is a strong function of the Kr* population. For example, the efficiency of forming Kr* is almost 60 percent when the fractional population is 10^{-5} and the electric field is 2 kV/cm · atm. This efficiency decreases to less than 10 percent when the fractional population is increased to 10^{-4}. The decrease in efficiency is caused by preferential cooling of the high energy tail of the electron distribution which produces Kr*, by the electron impact process $e + Kr^* \rightarrow Kr^{**} + e$. Fig. 15 is a plot of the average electron energy as a function of the fractional metastable population. The cooling effect is much stronger at smaller electric fields. The decrease in efficiency can be made up by increasing the electric field. However, the ionization rate (see Fig. 16) rapidly becomes so large that it precludes discharge stabilization by electron attachment (F_2) for cases where the discharge power exceeds the *e*-beam power into the laser medium (see following section).

Under controlled discharge conditions, one can choose the desired operating conditions and insure that the fractional metastable population is small enough so that the metastable production efficiency is high. In the case of the UV preionized discharge, however, the avalanching current and power densities result in an uncontrolled increase in excited state density. This in turn leads to a low metastable production efficiency and a relatively large absorption by excited and ionic states at the laser frequency. Hence the laser extraction efficiency is adversely affected.

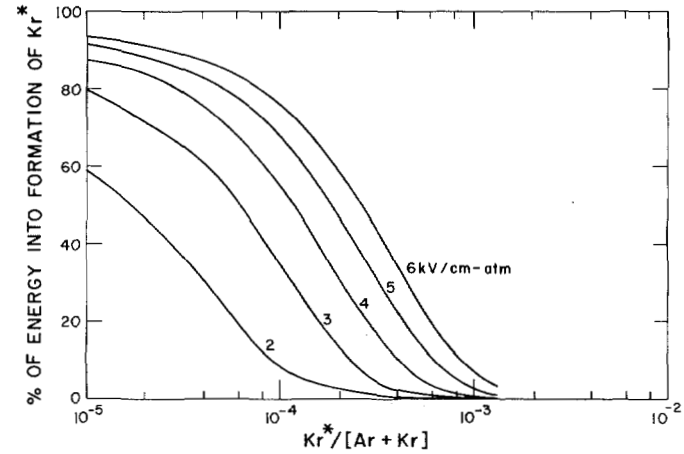


Fig. 14. Percentage of discharge power into Kr* as a function of fractional Kr* population for various applied electric fields.

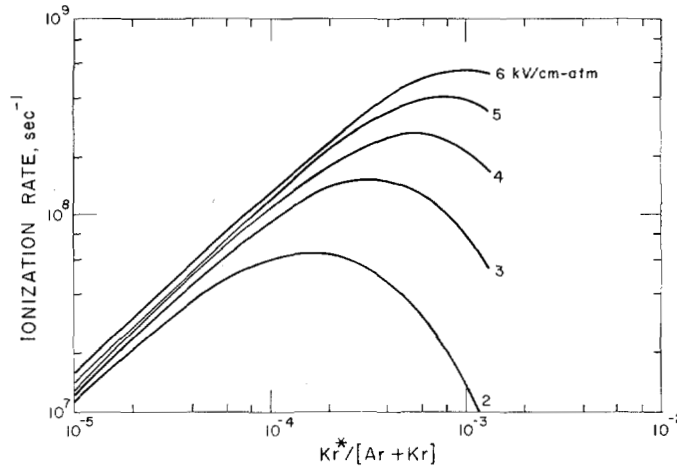


Fig. 15. Average electron energy as a function of fractional Kr* population.

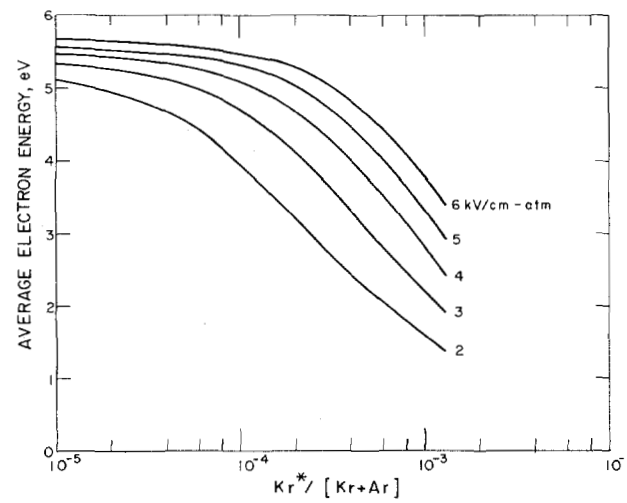


Fig. 16. Ionization rate as a function of fractional Kr* population.

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2) Discharge Stability

The discharge stability of these mixtures is also strongly affected by the rare gas excited states. To clarify this point let us evaluate the fractional metastable density under lasing conditions. Typically the input power density for efficient laser action is 100 J/l-μs. The discharge power that goes into producing metastables directly is given by

$$P_m = \eta_m \times 10^5 \text{ W/cm}^3 \tag{13}$$

where η_m is the metastable production efficiency. The metastable lifetime is determined primarily by halogen donors like F_2. In the presence of 0.2 percent F_2 at a mixture pressure of 2 atm, the rare gas metastable lifetime τ_d is about 15 ns. Assuming that η_m = 50 percent the steady-state metastable density M^* is about 5×10^{14} cm^{-3} and the fractional metastable population is 10^{-5}. For these conditions, the electron impact ionization rate of the metastables is 2×10^7 s^{-1} compared to 10^3 s^{-1} for the ground state. Hence, metastable ionization is the dominant ionization process for laser discharge conditions. The production and loss of discharge electrons n_e is given by

$$\frac{dn_e}{dt} = S + (\nu_i - \beta)n_e \tag{14}$$

where S is the production rate of electron/ion pairs by e-beam or UV ionization, ν_i is the ionization frequency by secondary electrons, and β is the attachment rate.

The production and loss of the metastables is given by

$$\frac{dM^*}{dt} = \langle \sigma v \rangle n_e M - \frac{M^*}{\tau_d} \tag{15}$$

where $\langle \sigma v \rangle$ is the rate constant for metastable production by electron impact and τ_d is the metastable lifetime. As we have discussed above, the dominant ionization is metastable ionization, so we can write

$$\nu_i = \langle \sigma v \rangle_i M^*. \tag{16}$$

Equations (14) and (15) are a pair of nonlinear coupled differential equations in n_e and ν_i. By a perturbation analysis it can be shown [53] that these equations predict a nonzero steady-state solution for n_e if [54]

$$\beta \geqslant 2\nu_{i0} \tag{17}$$

where ν_{i0} is the equilibrium ionization rate. From (14) it becomes clear that this stability criterion can be satisfied only if $S \neq 0$. Therefore it is clear that UV or e-beam preionized laser discharge will not be stable because the electron production occurs prior to the discharge pulse.

A system of nonlinear equations similar to (14) and (15) has been solved numerically for an e-beam current density of about 2 A/cm^2 at 150 keV [53]. In this analysis, we have also included Penning ionization and ionization of the ground state atoms. Fig. 17 shows the results of such an analysis. On the left-hand side, we have the stable discharge condition, i.e., the attachment rate is slightly greater than twice the equilibrium ionization rate. Notice that the discharge current reaches a constant value asymptotically. Another important feature for the stable discharge case is that the metastable production efficiency η_m remains above 75 percent. If we keep everything constant but decrease the attachment rate by 20 percent, we observe that the ionization rate increases and after 70 ns becomes greater than the attachment rate. For this case the discharge current increases faster than exponentially in time and the metastable production efficiency falls steeply. In Fig. 18 we see experimental current, voltage, and fluorescence traces for a KrF e-beam controlled discharge under stable and unstable operating conditions. The discharge cavity was filled to 2 atm with 0.1 percent F_2, 2 percent Kr, and 97.9 percent Ar. Stable discharge conditions were achieved when the 0.3 μF capacitor was charged to 8 kV. Notice that the discharge current reaches a steady-state value before the end of the e-beam current pulse. When the capacitor was charged to 10 kV the discharge was unstable. The discharge current increases slowly at first and has a shape very similar to the unstable case shown in Fig. 17. The discharge current is volumetrically unstable before it arcs. This assertion is verified by observing in Fig. 18, that the fluorescence is increasing for 40 ns subsequent to the onset of rapid current growth.

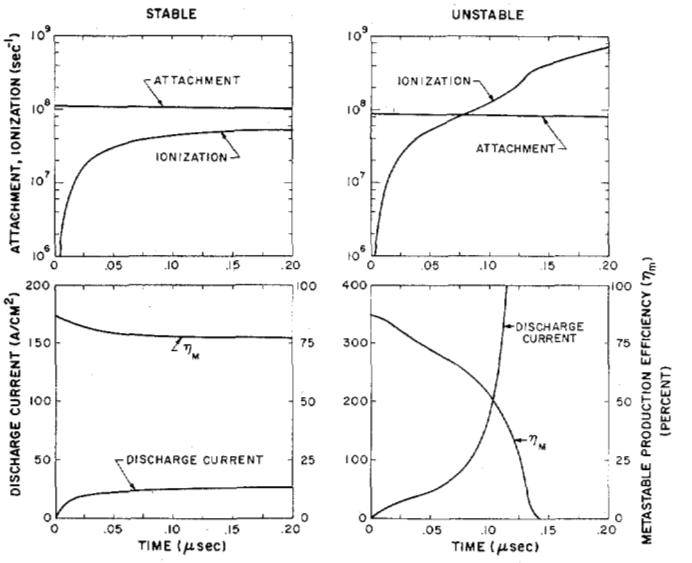


Fig. 17. Numerical analysis of a system of equations similar to (16) and (17).

In a UV preionized discharge, the electrons freed by photoionization rapidly form negative halogen ions

$$e + X_2 \longrightarrow X^- + X. \tag{18}$$

The negative ions then provide an easily ionizable source for production of discharge electrons [55]. Electron detachment presumably occurs by discharge-electron impact. The negative ions which are lost by recombination during the delay period between the UV preionizing pulse and the main discharge pulse, result in a more uniform spatial distribution. When the discharge electric field is applied the electron density avalanches rapidly by several orders of magnitude, and large discharge-energy inputs are possible.

3) Discharge Enhancement

In an e-beam controlled discharge one is interested in both discharge stability and large ($\geqslant$5) discharge enhancement. The discharge enhancement is defined here as the ratio between the

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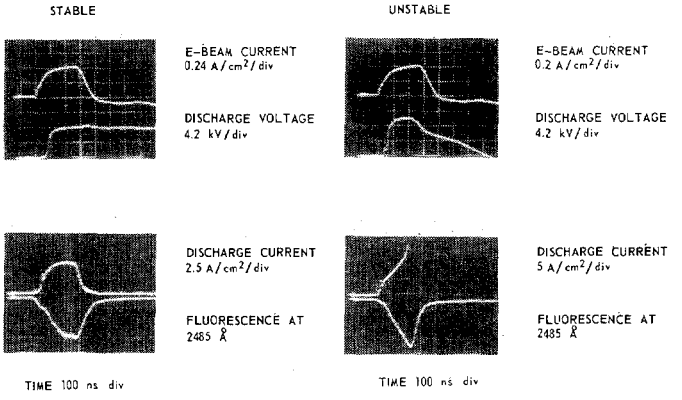


Fig. 18. Experimental results showing stable and unstable discharge conditions.

power deposited in the laser mixture by the discharge, P_d, to the power deposited by the e-beam P_{eb}. This ratio is given by

$$\frac{P_d}{P_{eb}} = \frac{eV_D E}{(\beta - \nu_{i0})E_i} \tag{19}$$

where V_D is the average discharge electron drift velocity, E is the applied electric field, and E_i is the energy required to create an electron-ion pair by the beam electrons. For discharge stability, we have shown that $\beta \geqslant 2\nu_{i0}$, so for an e-beam stabilized discharge, (19) can be rewritten as

$$\frac{P_d}{P_{ed}} \leqslant \frac{eV_D E}{\nu_{i0} E_i}. \tag{20}$$

Using the curves in Figs. 15 and 16, the ionization rate can be expressed as a function of the metastable production efficiency η_m for a given electric field. Thus we can plot the discharge enhancement ratio as a function of the metastable production efficiency. Fig. 19 shows such a plot for an electric field of 3 kV/cm-atm. The analysis predicts that one can obtain a stable discharge with a large enhancement ($\geqslant 5$) and a large metastable production efficiency (70–80 percent). However, this is obtained at a price, i.e., the power into the laser mixture decreases as one moves along the curve from left to right. In fact, the maximum metastable efficiency is approached asymptotically as the discharge power, and hence metastable density, goes to zero. It appears reasonable however to obtain discharge enhancement factors of 5–10 and metastable pumping efficiencies of 70–80 percent at a discharge power input of 100 J/l-atm-μs.

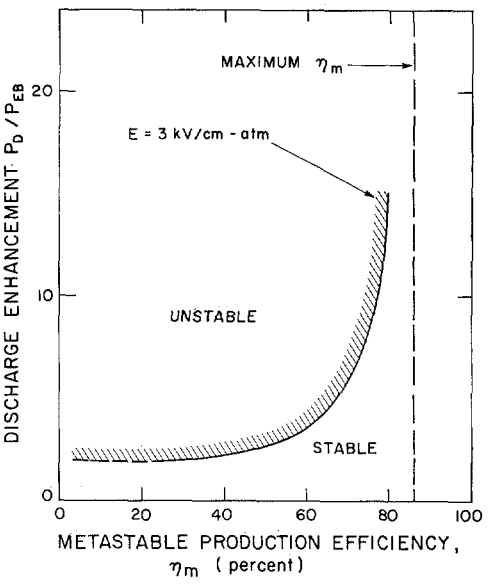


Fig. 19. Plot of discharge enhancement as a function of metastable production efficiency at the stable discharge limit.

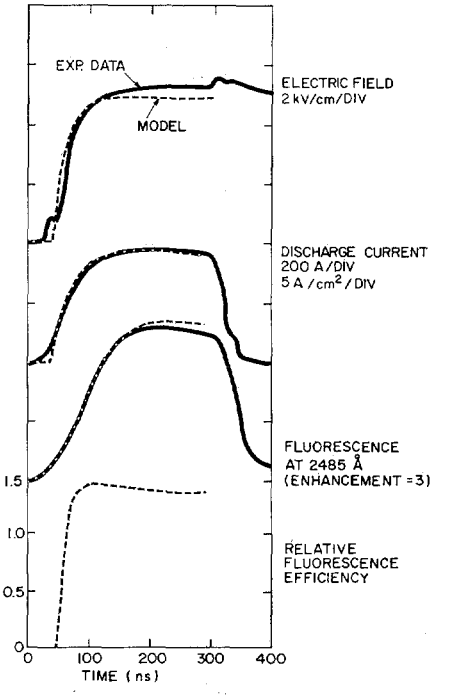


Fig. 20. Measured and predicted fluorescence when the capacitor is charged to 10 kV. The discharge cavity contained 2 atm of 93.7 percent Ar, 6 percent Kr and 0.3 percent F_2.

C. Kinetic Discharge Model

Using the rate constants predicted by the Boltzmann code, we have developed a self-consistent code that follows the temporal evolution of the secondary electrons, positive and negative ions, Ar*, Kr*, and KrF*, etc. We couple our code to a simultaneous set of differential equations that describe the electrical circuit. The outputs of this code include the temporal evolution of the discharge current and voltage and the KrF* fluorescence for a given ionization level, discharge capacitor charge voltage, and gas mixture. The predictions of this discharge model have been compared with our KrF laser discharge experiments. The cavity was filled with a 2 atm mixture of 93.7 percent Ar, 6 percent Kr, and 0.3 percent F_2. Fig. 20 shows the experimental results and theoretical predictions when the 0.3 μF capacitor is charged to 10 kV. The top trace is the discharge voltage. The second trace is the discharge current. The third trace is the KrF* fluorescence. By the end of the pulse the enhancement in the fluorescence is 3. The KrF* are being produced with a maximum efficiency of 1.4 times the efficiency of metastable production by a pure e-beam. Fig. 21 shows the results when the capacitor is charged to 16 kV. In this case the discharge current continually increases until the discharge goes through the glow to arc transition which is marked by an abrupt decrease in KrF* fluorescence. We believe that the initial (slow) increase in the discharge current is caused by a

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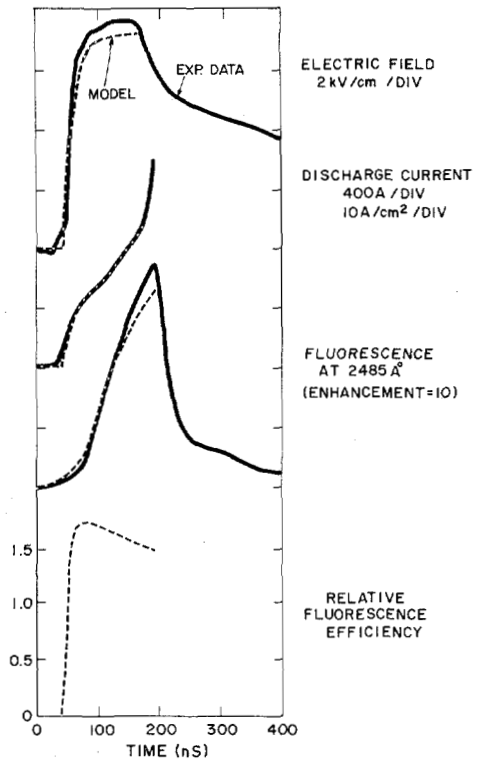


Fig. 21. Measured and predicted fluorescence when the capacitor is charged to 16 kV. The discharge cavity contained 2 atm of 93.7 percent Ar, 6 percent Kr, and 0.3 percent F_2.

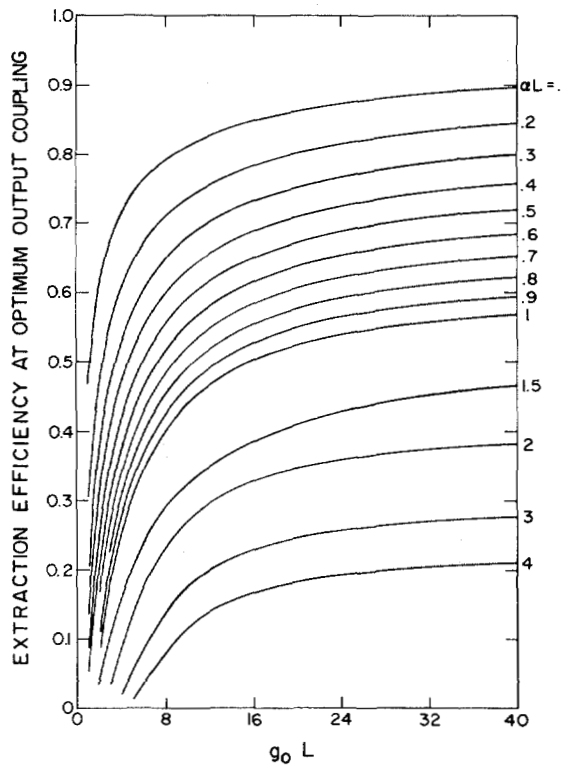


Fig. 22. Extraction efficiency at optimum output coupling as a function of g_0L, and αL.

volumetric discharge instability discussed previously. The efficiency for producing the KrF* rises rapidly to 1.7 times the efficiency of producing metastables in a pure *e*-beam and then begins to fall despite the fact that the voltage is constant. The KrF* production efficiency decreases because as the metastable density increases, the discharge pumping efficiency of Ar* and Kr* falls.

V. Power Extraction

The power extractable from a laser cavity was analyzed by Rigrod [56]. In his analysis he assumed a stable optical resonator, a homogeneously broadened gain coefficient, and a plane electromagnetic wave solution. He also assumed steady-state conditions and included mirror losses. The Rigrod model has been extended to account for a uniformly distributed absorption in the active medium [57]. Such a modification is necessary because of the ground, excited, and ionic state absorption at the laser wavelengths as will be discussed subsequently. For this case the extraction efficiency can be presented as a function of the small-signal gain g_0, absorption coefficient α, the laser length L, the saturation flux ϕ_{sat}, and the reflectivity of the mirrors. The steady-state analysis is valid provided the laser pulse is much longer than the gain rise time and photon transit time through the laser cavity. So this analysis is applicable for the steady-state laser pulses ($\sim 1\ \mu s$) obtained by *e*-beam and *e*-beam controlled discharge pumping. The analysis is not valid for the short pulses typical of the UV preionized lasers.

*A. Power Extraction in KrF**

Fig. 22 shows the variation of the optimum extraction efficiency as a function of the small-signal gain per pass g_0L for various intrinsic nonsaturable absorptions per pass, αL. Since part of the absorption might be saturable, the αL used should be evaluated in the presence of cavity flux. The efficiency is calculated for the optimum mirror reflectivity (output coupling) and it is assumed that the lower laser level is repulsive (as in KrF*). An important feature of these curves is that a laser that has intrinsic absorption cannot be scaled efficiently in length indefinitely. For example, for a g_0L of 8 and an αL of 0.5, the extraction efficiency is 55 percent. If the length is increased by a factor of 5 so g_0L is 40 and αL is 2.5, the extraction efficiency drops to 33 percent.

Absorption in the KrF* active medium has been investigated [58]. There are a number of species that photoabsorb at 249 nm. Steunenberg and Vogel [59] have determined that the F_2 photoabsorption cross section is about 1.3×10^{-20} cm^2. By probing an *e*-beam excited mixture with a flashlamp pumped dye laser, broadband absorption has been observed [58]. This absorption can be explained as contributions from positive and negative ionic states (Ar_2^+, Kr_2^+, F^-), by triatomic species such as Kr_2F^*, and by the high lying rare gas electronic states Ar** (4*p* states). Table IV lists the dominant absorbing species and their respective cross sections at 249 nm (other than F_2). This table also gives the number densities of these species as predicted from our kinetic code for 1.5 and 6 A/cm^2 *e*-beam current densities.

The F^- photodetachment cross section has been measured by Mandl [60] between 245 and 270 nm. The photoionization cross sections of Ar** and Kr** (4*p* and 5*p* states, respectively) have been calculated by Hyman [61]. Stevens, Gardner, and Karo [62] and Wadt and Hay [63] have recently calculated the Ar_2^+ photodissociation cross section ($^2\Sigma_u^+ \rightarrow {}^2\Sigma_g^+$). The Kr_2^+ photoabsorption cross section was estimated by red shift-

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TABLE IV
KrF* PHOTOABSORPTION

SPECIE	PHOTOABSORPTION CROSS SECTION cm^2	SOURCE	NUMBER DENSITY cm^{-3}		ABSORPTION cm^{-1}	
			1.5 A/cm^2	6 A/cm^2	1.5 A/cm^2	6 A/cm^2
F^-	5.6×10^{-18}	MANDL	1.1×10^{14}	2.2×10^{14}	6.2×10^{-4}	1.2×10^{-3}
Kr_2^+	1.6×10^{-18}	ESTIMATED	4.3×10^{13}	4.9×10^{13}	6.9×10^{-5}	7.9×10^{-5}
Kr_2F^*	1.6×10^{-18}	ESTIMATED	3.7×10^{14}	1.5×10^{15}	5.9×10^{-4}	2.4×10^{-3}
Ar_2^+	1.3×10^{-17}	STEVENS ET AL	5.3×10^{12}	1.8×10^{13}	6.9×10^{-5}	2.3×10^{-4}
Ar**,Kr**	6×10^{-18}	HYMAN	6.2×10^{12}	2.6×10^{13}	3.7×10^{-5}	1.6×10^{-4}
TOTAL CALCULATED ABSORPTION					1.4×10^{-3}	4.1×10^{-3}
TOTAL MEASURED ABSORPTION					1.5×10^{-3}	4.4×10^{-3}

ing the Ar_2^+ cross section as given by Stevens, Gardner, and Karo [62] by 20 nm. Finally we have assumed that Kr_2F^* will have the same absorption cross section as Kr_2^+ [64]. The total absorption coefficient measured at an *e*-beam current density of 1.5 A/cm^2 is 1.5×10^{-3} cm^{-1}. From the data listed in Table IV, an absorption coefficient of 1.4×10^{-3} cm^{-1} was calculated. At 6 A/cm^2 the measured and calculated values were 4.4×10^{-3} cm^{-1} and 4.1×10^{-3} cm^{-1}, respectively.

The gain on line center has also been measured. From the gain and absorption measurements, the stimulated emission cross section σ_s is 2.5 ± 0.3 Å^2. Tellingheusen *et al.* [65] have analyzed the KrF* spontaneous emission spectrum and evaluated σ_s to be 2.6 ± 0.3 Å^2. From these results one can conclude that the self-absorption of the KrF* radiation to the Rydberg levels is negligible.

The above results have been used to predict the performance of a 1 m KrF* laser device. The best results from this device with direct *e*-beam pumping are listed in Table V. For these conditions, $g_0L = 3.7$, $\alpha L \approx 0.35$, so that the predicted extraction efficiency is 46 percent. This corresponds to a computed intrinsic efficiency of 9.3 percent compared to the measured efficiency of 9 percent. The optical energy extracted was 102 J in 8.5 l corresponding to a specific output energy of 12 J/l.

Finally, the best laser results obtained by *e*-beam controlled discharge pumping are summarized in Table VI. These results are limited by the fact that the discharge is unstable for electric fields greater than 2 kV/cm · atm. Ideally, it would be preferable to operate at electric fields of 3 kV/cm · atm. At such electric field strengths, both the discharge enhancement and intrinsic efficiency would be higher. Theoretically the intrinsic efficiency should increase to 15 percent and the enhancement to 4. The discharge instability is probably the result of nonuniformities in the ionization produced by the beam electrons. For a stable discharge, it is necessary to satisfy the stability criterion in the region where input power is largest; hence, a nonuniform discharge will not be as efficient as a uniform discharge.

TABLE V
PERFORMANCE: E-BEAM PUMPED KrF LASER (0.2 PERCENT F_2/4 PERCENT Kr/95.8 PERCENT Ar AT 1.7 atm)

●	LASER PERFORMANCE		
	– LASER ENERGY	=	102 JOULES
	– ACTIVE VOLUME	=	8.5 LITERS
	– LASER ENERGY DENSITY	=	12 J/LITER
	– LASER ENERGY/ENERGY DEPOSITED	=	9%
	– LASER PULSE LENGTH	=	600 nsec
●	E-BEAM CHARACTERISTICS		
	– BEAM CURRENT DENSITY	=	11.5 A/cm^2
	– BEAM ENERGY	=	250 keV

*B. Power Extraction in XeF**

The best XeF* laser performance reported to date in a 1 m device with *e*-beam pumping is listed in Table VI. These results were obtained in a Ne-rich mixture. The experimental efficiency of 2.6 percent is less than one-third that of KrF*. In part the decreased efficiency is caused by the lower quantum efficiency. In Ne-rich mixtures the longer relaxation time of the upper manifold and the bound lower level decrease the maximum obtainable efficiency. In Ar-rich mixtures additional quenching and active medium absorption decrease the extraction efficiency.

The dominant absorbing species in *e*-beam pumped XeF* mixtures have been identified by systematic absorption measurements [66]. The measured and calculated absorption cross sections of the different species are listed in Table VII. In Fig. 23 a comparison of the absorption in Ar and

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TABLE VI
E-BEAM CONTROLLED DISCHARGE PUMPING OF KrF* 1 m DEVICE
(0.5 PERCENT F_2/10 PERCENT Kr/89.5 PERCENT Ar AT 1.5 atm)

• LASER PERFORMANCE		
– LASER ENERGY	=	75 JOULES
– ACTIVE VOLUME	=	7.5 LITERS
– LASER ENERGY DENSITY	=	10 J/LITER
– LASER ENERGY/ENERGY DEPOSITED	=	9.5%
– LASER PULSE LENGTH	=	500 nsec
– OUTPUT COUPLING	=	71%
• DISCHARGE CHARACTERISTICS		
– DISCHARGE ELECTRIC FIELD	≃	2 kV/cm–atm
– DISCHARGE CURRENT DENSITY	=	70 A/cm^2
– DISCHARGE ENERGY DEPOSITED	=	70 J/LITER
• E-BEAM CHARACTERISTICS		
– BEAM CURRENT DENSITY	=	4 A/cm^2
– BEAM ENERGY	=	300 keV
– BEAM ENERGY DEPOSITED	=	35 J/LITER

Ne-rich mixtures is shown. These mixtures contained 0.5 percent Xe and 0.2 percent F_2. From this figure it is apparent that in $Ne/Xe/F_2$ mixtures the observed photoabsorption is much less than in $Ar/Xe/F_2$ mixtures [67]. There are two main reasons for the decreased photoabsorption in Ne-rich mixtures at XeF wavelengths. 1) Ne_2^+ photoabsorption at XeF wavelengths is much smaller than Ar_2^+ photoabsorption [63], [66], and 2) Xe_2^+ formation is much slower because the intermediate complex $NeXe^+$, in contrast to $ArXe^+$, is probably very weakly bound or unbound [30].

Fig. 24 shows the bound upper and lower XeF laser levels. At 300 K, the dominant laser transition is the $v' = 0 \rightarrow v'' = 3$ (353 nm). The extraction efficiency will depend on the lower level lifetime and the vibrational relaxation rate of the upper manifold. A detailed analysis starting with rate equations for the upper and lower levels shows that the extraction efficiency can be written as [68]

$$\eta_{ext} = \eta \left(\frac{\theta \tau_u / \tau_v}{1 + \theta \tau_u / \tau_v} \right) \lim_{\phi \rightarrow \infty} \left[1 - \frac{N^*(v=0)}{N_0^*(v=0)} \right] \tag{21}$$

where η is the extraction efficiency plotted in Fig. 22. The second term accounts for the finite vibrational relaxation time of the upper laser manifold. θ is the fractional population in $v = 0$ assuming vibrational equilibrium, τ_u is the upper level lifetime, and τ_v is the vibrational relaxation time. When $\theta \tau_u / \tau_v \gg 1$ this term approaches unity. The last term accounts for the inefficiency resulting from the finite lifetime of the lower laser level ($v'' = 3$). $N^*(v=0)$ is the population density of the zeroth vibrational level of the upper manifold, in the presence of a cavity flux ϕ, and N_0^* is the same population in the absence of flux ($\phi = 0$). For a repulsive lower level, as in the case of KrF*, the third term is unity. Physically, this last term must be evaluated in the limit $\phi \rightarrow \infty$; otherwise, it accounts again for inefficiencies caused by collisional quenching and spontaneous radiation already included in η. By taking this limit, only the inefficiency due to bottlenecking is retained. This term is written in the form shown for con-

TABLE VII
COMPARISON OF THE EXPERIMENTAL AND THEORETICAL ABSORPTION COEFFICIENTS OF MOLECULAR RARE GAS IONS AT 340 nm

SPECIES	σ cm^2 300°K EXP.	THEOR.
Ar_2^+	1.7×10^{-17}	2×10^{-17} (REF 62); 8.7×10^{-18} (REF 63)
Xe_2^+	1.2×10^{-16}	4.8×10^{-17} (REF 63)
Ne_2^+	–	9.6×10^{-19} (REF 63)

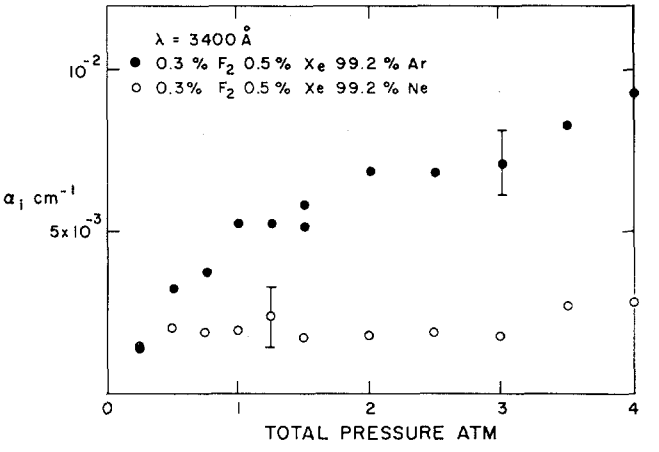


Fig. 23. Excited state absorption in $Ar/Xe/F_2$ mixtures.

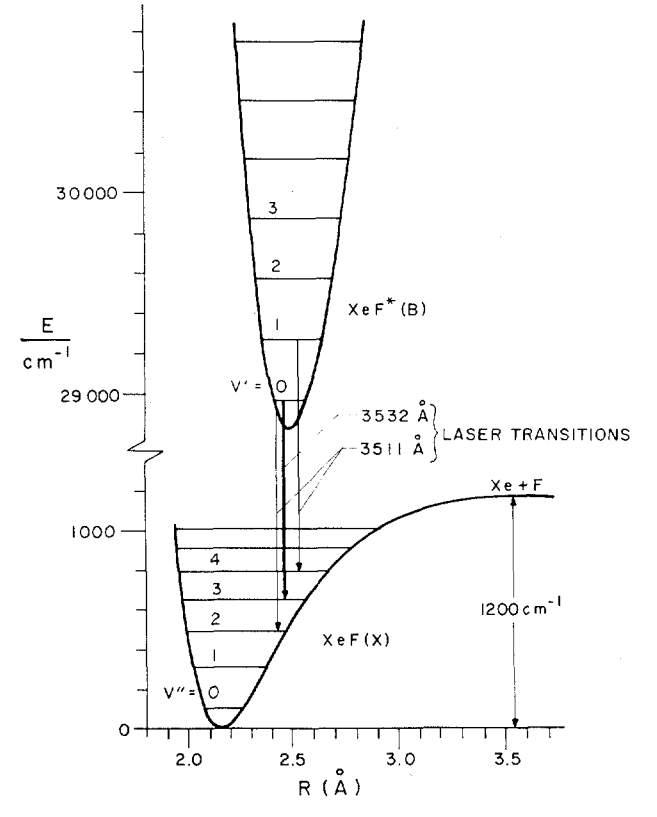


Fig. 24. Schematic showing the upper and lower levels of XeF*.

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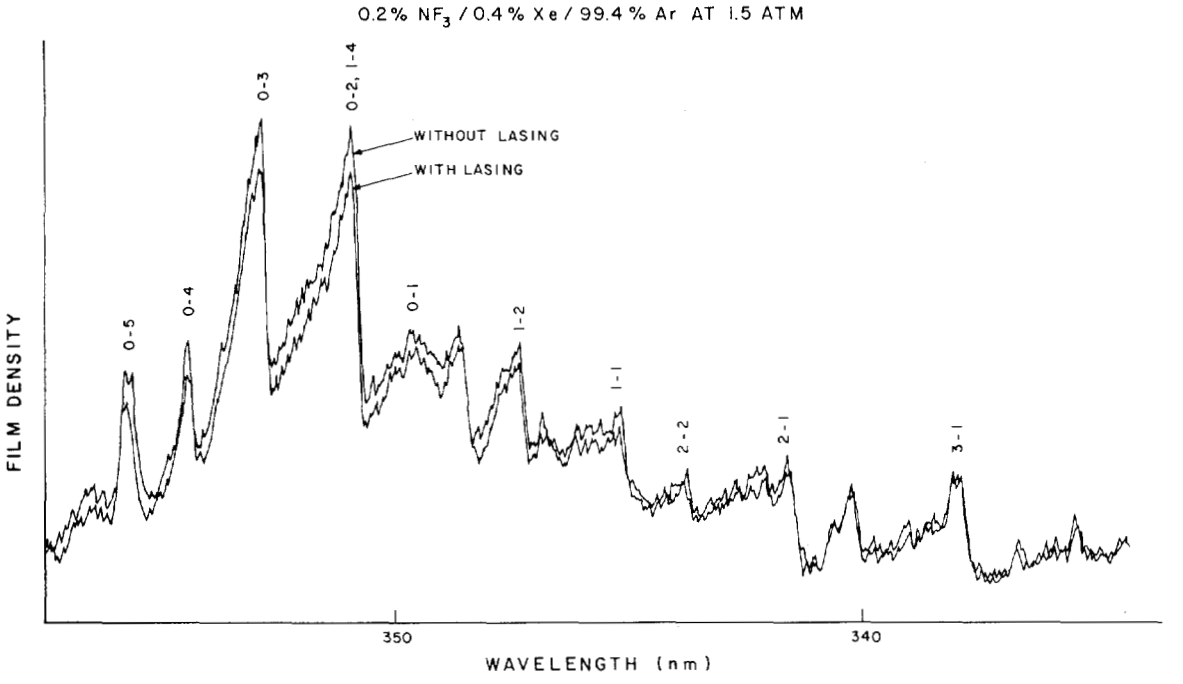


Fig. 25. XeF* sidelight spectra with and without lasing (Ar diluent).

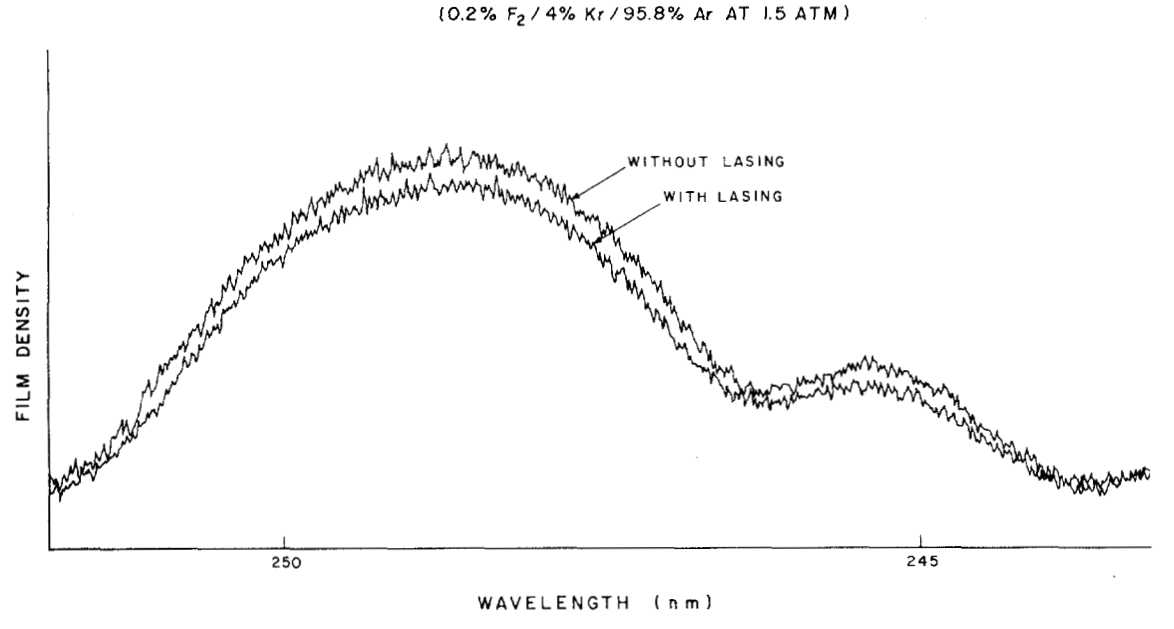


Fig. 26. KrF* sidelight spectra with and without lasing.

venience in isolating and evaluating the losses caused by bottlenecking from experimental data. This term is independent of ϕ and can also be written in terms of the radiative lifetime, collisional quenching time, and lower state lifetime [68].

We will first consider the effect of the vibrational relaxation of the upper manifold. In Ar-rich mixtures, as we have discussed previously, the vibrational relaxation is faster than in Ne-diluted mixtures. The second term in (21) is almost unity when Ar is used as a buffer. The same is true in KrF laser mixtures with an Ar buffer. Figs. 25 and 26 show the time integrated sidelight spectra of XeF and KrF, with Ar buffer, for lasing and nonlasing conditions. From these figures it is clear that the vibrational relaxation is rapid enough and practically all the vibrational levels are well coupled to the lasing level. Fig. 27 shows time-integrated sidelight spectra of XeF in a $Ne/Xe/NF_3$ mixture for lasing and nonlasing conditions. From this figure it is evident that the laser radiation depresses only the $v' = 0$ population density while the higher vibrational levels are not affected. As a result, the second term in (21) is about 0.6, i.e., 60 percent of the available energy can be extracted.

In contrast to KrF*, the lower level of XeF* is bound by about 0.15 eV. In steady state at ambient temperature and

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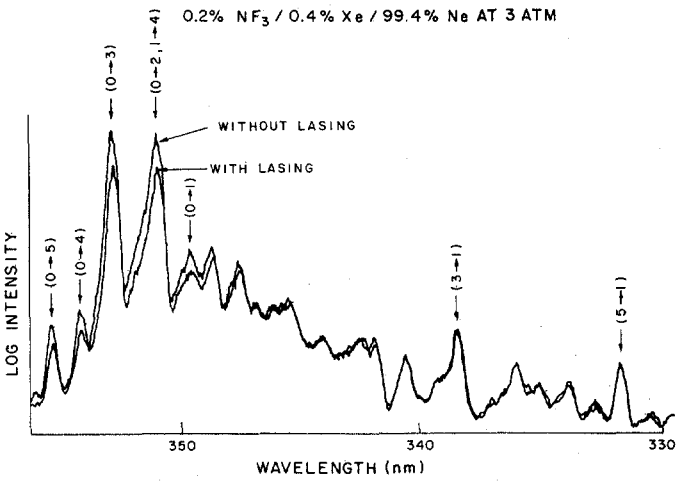


Fig. 27. XeF* sidelight spectra with and without lasing (Ne diluent).

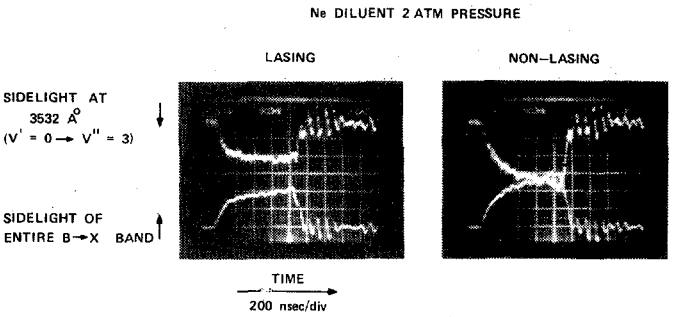


Fig. 28. XeF* sidelight emission at 353. 2 nm with and without lasing.

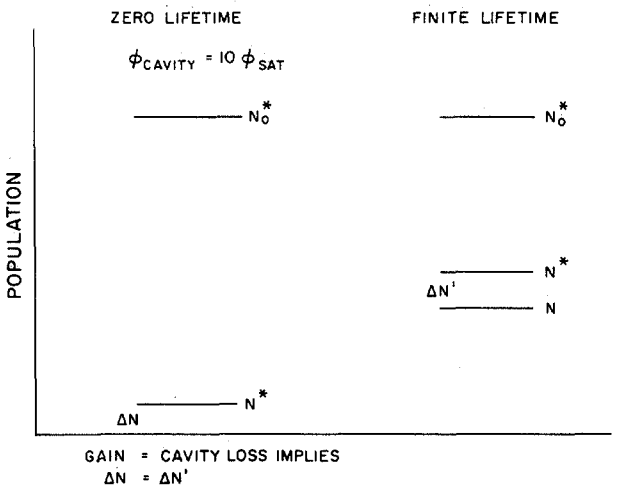


Fig. 29. Schematic presentation of the effect of the finite lifetime of the lower laser level on the extraction efficiency.

typical laser mixture pressures, most of the XeF will dissociate into Xe + F. However, if the dissociation rate τ_D is comparable to the upper level lifetime, the extraction efficiency will be decreased by the third term of (21). Fig. 28 shows the sidelight emission at 353.2 nm ($v' = 0 \rightarrow v'' = 3$) under lasing and nonlasing conditions. Under lasing conditions, we ensured that the cavity flux was at least an order of magnitude larger than the saturation flux. From the sidelight fluorescence the ratio of $N^*(v = 0)/N_0^*(v = 0)$ in the limit $\phi/\phi_s \gg 1$ is determined to be ≈ 0.6. Therefore, the last term of (21) is 0.4. The decrease in the extraction efficiency due to the lower level can be understood from the schematic diagram shown in Fig. 29. For a repulsive lower level if $\phi/\phi_s = 10$, the sidelight will be depressed by a factor of 11. For this condition, a finite $\Delta N = (N^* - N)$ is required to overcome the cavity loss. In the case of a repulsive lower level $N = 0$, hence $N^* = \Delta N$. For a bound lower level and steady-state lasing conditions, the population of the ground state will be finite and dependent on the ground state lifetime. For the same cavity loss (same ΔN) the N^* needed to sustain lasing action is increased ($N^* = N + \Delta N$). Hence the extraction efficiency is lower by the factor given in (21).

VI. Conclusion

The rare gas fluoride lasers can be very efficient and have the potential of being scaled to high average power. To date, the best efficiencies and single pulse energies have been obtained from KrF* which has a repulsive lower level. Formation and quenching processes and laser power extraction in KrF* lasers are understood well enough to enable laser performance predictions. In the XeF* laser the chemistry and physics are more complicated. It appears that the efficiency is constrained by the bound lower level, the finite vibrational relaxation of the upper level, and intrinsic active medium absorption. The intrinsic absorption and quenching are greatly reduced by using Ne as a buffer instead of Ar; however, in the presence of Ne, the vibrational relaxation of the upper laser level is slower than in the presence of Ar.

New families of exciplex lasers having ionic upper levels, as in the rare gas fluorides, can be researched. The mercury monohalides discovered recently are examples of such a class of lasers. As in the case of the rare gas halides, these exciplexes were first lased by pure *e*-beam pumping [6]. Laser action has also been obtained by *e*-beam controlled discharge pumping [69]. The mercury monohalide lasers radiate in the visible and have the potential of achieving high power and high efficiencies. There are also triatomic exciplexes analogous to the rare gas halides that could possibly lase. These systems will be far more complex with a high density of states and numerous curve crossings and hence the probability of achieving efficient lasing in the triatomics is expected to be smaller than in the diatomic exciplexes.

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High Stability CW Waveguide CO_2 Laser for High Resolution Saturation Spectroscopy

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Abstract—**We describe a stable waveguide CO_2 laser for spectroscopic applications. The short term spectral width is less than 10 kHz and a 10^{-12} flicker floor is reached for the Allan variance. A study of the oscillation frequency versus resonator length demonstrates the major influence of the anomalous index of refraction of the medium. Saturation resonances with a 20 kHz peak-to-peak width have been obtained with a very good signal-to-noise ratio. A complete spectrum of SF_6 for P (16) at 947.74 cm^{-1} is given as an example to illustrate the 500 MHz bandwidth of the laser.**

INTRODUCTION

THE extension of ultrahigh resolution saturation spectroscopy [1] to all corners of the spectrum is only limited by the slow development of tunable laser sources of high enough quality. Practically, a few milliwatts in the lowest order spatial mode are necessary with the best possible short-term frequency stability. In the 10 μm region the various solid-state devices have apparently not yet been able to meet these requirements and this situation has stimulated a considerable interest among the spectroscopists for the development of high pressure waveguide CO_2 lasers [2]–[4]. Spectra of various molecules have already been recorded with such lasers used in Doppler-limited linear spectroscopy [5]–[8]. We have built a very stable waveguide CO_2 laser to be used in our high resolution saturation spectrometer. The design and performances of this source are described in this paper. We display a saturation spectrum of SF_6 to illustrate the resolving and tuning capabilities of the spectrometer and discuss present limitations and further improvements.

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